A Conditional Landauer–Nelson Bridge with Non-Markovian TICE Closure

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We prove that five conditions are pairwise equivalent for a quantum Brownian particle coupled to a thermal bath: (i) the thermal-to-quantum diffusion ratio is $D_{\rm th}/D_{\rm q} = 2$; (ii) the friction coefficient is $\gamma_0 = k_{\rm B}T/\hbar$; (iii) the bath correlation time is $\tau_{\rm cor} = \hbar/(k_{\rm B}T)$; (iv) the bridge energy $E_{\rm bridge} \equiv k_{\rm B}T\ln(D_{\rm th}/D_{\rm q}) = k_{\rm B}T\ln 2$ equals one Landauer bit of erasure; (v) the ratio $D_{\rm th}/D_{\rm q}$ is temperature-independent. Exact deviation formulas $\Delta E = -k_{\rm B}T\ln(1+\delta)$ are derived for departures from this closure. We embed the result in a non-Markovian framework by introducing a TICE closure factor $C_{\rm TICE}(t) = k_{\rm B}T/[\hbar\,\gamma_{\rm eff}(t)]$ that generalizes the theorem to memory-kernel–dependent effective friction, yielding $E_{\rm bridge}^{\rm TICE} = k_{\rm B}T\ln(2C_{\rm TICE})$. The theorem is conditional on one postulate—that thermal diffusion accounts for both time-directed noise channels in Nelson's stochastic mechanics—which we motivate from five independent physical contexts but do not derive from first principles.

I. INTRODUCTION

Landauer's principle [?] bounds the minimum cost of irreversible bit erasure at $E_{\rm Landauer} = k_{\rm B}T\ln 2$. Nelson's stochastic mechanics [?] ascribes to every quantum particle a diffusion coefficient $D_{\rm q} = \hbar/(2m)$. The Caldeira–Leggett model [?] gives a thermalized particle the diffusion $D_{\rm th} = k_{\rm B}T/(m\gamma_0)$ via the fluctuation–dissipation theorem.

In this Letter we show that a single closure condition—$D_{\rm th} = 2D_{\rm q}$—locks the friction to $\gamma_0 = k_{\rm B}T/\hbar$ and produces a bridge energy of exactly one Landauer bit. We state the theorem, prove it, derive exact deviation formulas, embed it in a memory-kernel framework, give one explicit kernel solution, and identify three falsification protocols. We are explicit throughout about what is proven and what is postulated.

II. DEFINITIONS

Consider a quantum Brownian particle of mass m coupled to a heat bath at temperature T through an Ohmic–Drude spectral density $J(\omega) = m\gamma_0\omega\,\Omega^2/(\omega^2+\Omega^2)$, where Ω is the Drude cutoff [?]. In the high-T Markov regime ($k_{\rm B}T \gg \hbar\Omega$) the classical FDT gives $D_{\rm th} = k_{\rm B}T/(m\gamma_0)$.

Define the *diffusion ratio* and *bridge energy*:

$$R \equiv \frac{D_{\rm th}}{D_{\rm q}} = \frac{2k_{\rm B}T}{\gamma_0\hbar}, \tag{1}$$

$$E_{\rm bridge} \equiv k_{\rm B}T\ln R = k_{\rm B}T\ln\!\left(\frac{2k_{\rm B}T}{\gamma_0\hbar}\right). \tag{2}$$

Both are manifestly independent of m.

III. EQUIVALENCE THEOREM

Theorem. *The following five conditions are pairwise equivalent:*

(i) $D_{\rm th} = 2D_{\rm q}$.

(ii) $\gamma_0 = k_{\rm B}T/\hbar$.

(iii) $\tau_{\rm cor} \equiv 1/\gamma_0 = \hbar/(k_{\rm B}T)$.

(iv) $E_{\rm bridge} = k_{\rm B}T\ln 2 = E_{\rm Landauer}$.

(v) $\partial R/\partial T = 0$ at fixed R.

Proof. (i)⇔(ii): $k_{\rm B}T/(m\gamma_0) = \hbar/m$ gives $\gamma_0 = k_{\rm B}T/\hbar$. (ii)⇔(iii): definition. (ii)⇒(iv): $E_{\rm bridge} = k_{\rm B}T\ln[2k_{\rm B}T/(k_{\rm B}T)] = k_{\rm B}T\ln 2$. (iv)⇒(ii): $k_{\rm B}T\ln[2k_{\rm B}T/(\gamma_0\hbar)] = k_{\rm B}T\ln 2$ implies $\gamma_0\hbar = k_{\rm B}T$. (v)⇔(ii): $R = 2k_{\rm B}T/(\gamma_0\hbar)$ is T-independent iff $\gamma_0 \propto T$; combined with $R = 2$ this gives $\gamma_0 = k_{\rm B}T/\hbar$ uniquely. □

The content of (iv) is that the unique T-independent closure produces a bridge energy of *exactly one Landauer bit*—unit information erasure cost.

IV. EXACT DEVIATION FORMULA

Write $\gamma_0 = \gamma^*(1+\delta)$ with $\gamma^* \equiv k_{\rm B}T/\hbar$. Then:

$$R = \frac{2}{1+\delta}, \qquad \Delta E \equiv E_{\rm bridge} - E_{\rm Landauer} = -k_{\rm B}T\ln(1+\delta). \tag{3}$$

This is *exact*. The sign encodes thermodynamics: $\delta > 0$ (overdamped) ⇒ $\Delta E < 0$ (sub-Landauer); $\delta < 0$ (underdamped) ⇒ $\Delta E > 0$ (super-Landauer). The second law ($E_{\rm bridge} \geq E_{\rm Landauer}$ for irreversible erasure [? ?]) requires $\delta \leq 0$ at equilibrium.

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V. CLOSURE CONDITION AND CHAIN OF INFERENCE

The theorem rests on one postulate atop four layers of established physics:

Layer 1 (Caldeira–Leggett [?]): Ohmic–Drude bath $\Rightarrow$ noise kernel $N(\tau) \simeq (mk_{\rm B}T\gamma_0\Omega/\hbar)\, e^{-\Omega|\tau|}$.

Layer 2 (KMS $\to$ FDT): $\tilde{N}(\omega) = \coth(\hbar\omega/2k_{\rm B}T)\,\tilde{D}(\omega)$ $\Rightarrow N(\tau) = (k_{\rm B}T/\hbar)\,\eta(\tau)$ at high T.

Layer 3 (Green–Kubo): $D_{\rm th} = k_{\rm B}T/(m\gamma_0)$.

Layer 4 (Nelson [?]): $D_{\rm q} = \hbar/(2m)$.

Conjecture 1 (closure condition). *The thermal diffusion $D_{\rm th}$ accounts for both time-directed noise channels in Nelson's bidirectional stochastic mechanics, giving $D_{\rm th} = 2D_{\rm q}$.*

Under Conjecture 1 the theorem follows algebraically. The conjecture is *motivated*—not proven—by five observations: (a) Nelson's forward and backward SDEs contribute independent noise of equal variance, summing to $2D_{\rm q}$; (b) the KMS thermal time $\hbar/(k_{\rm B}T)$ coincides with $\tau_{\rm cor} = 1/\gamma^*$; (c) the Caldeira–Leggett high-T friction scales as $k_{\rm B}T/\hbar$; (d) at $R = 2$ the bridge energy equals exactly one Landauer bit—the minimum cost of erasing one bit of time-direction information; (e) $R = 2$ is the unique value that is simultaneously T-independent and matches Landauer.

VI. TICE CLOSURE FACTOR AND KERNEL EXAMPLE

A. Non-Markovian generalization

Define the TICE closure factor

$$C_{\rm TICE}(t) \equiv \frac{k_{\rm B}T}{\hbar\,\gamma_{\rm eff}(t)}, \qquad R(t) = 2\,C_{\rm TICE}(t), \tag{4}$$

where $\gamma_{\rm eff}(t)$ is the time-dependent effective friction from a memory kernel $K(\tau)$:

$$\gamma_{\rm eff}(t) = \gamma_0 + \beta\int_0^t K(t-\tau)\,|\psi(\tau)|^2\,d\tau. \tag{5}$$

The master formula for the bridge energy becomes:

$$\boxed{E^{\rm TICE}_{\rm bridge}(t) = k_{\rm B}T\ln\bigl(2\,C_{\rm TICE}(t)\bigr)} \tag{6}$$

$C_{\rm TICE} = 1$: exact Landauer. $C_{\rm TICE} > 1$: super-Landauer (thermodynamically allowed for reversible work extraction). $C_{\rm TICE} < 1$: sub-Landauer (forbidden by second law for irreversible erasure).

B. Ohmic–Drude kernel

For $K(\tau) = A\,e^{-\Omega|\tau|}$ with $A = mk_{\rm B}T\gamma_0\Omega/\hbar$ and coupling β:

$$\gamma_{\rm eff}(t) = \gamma_0 + \frac{\beta A}{\Omega}\bigl(1 - e^{-\Omega t}\bigr), \tag{7}$$

so that $C_{\rm TICE}(t) \to k_{\rm B}T/[\hbar(\gamma_0 + \beta A/\Omega)]$ as $t \to \infty$. The bridge energy converges to $E^{\infty}_{\rm bridge} = k_{\rm B}T\ln[2k_{\rm B}T/(\hbar\gamma_0 + \hbar\beta A/\Omega)]$, and the crossing time t^* at which $E^{\rm TICE}_{\rm bridge}(t^*) = k_{\rm B}T\ln 2$ is:

$$t^* = \frac{1}{\Omega}\ln\!\left(\frac{\beta A/\Omega}{\gamma_0 + \beta A/\Omega - k_{\rm B}T/\hbar}\right), \tag{8}$$

real and positive when $\gamma_0 < k_{\rm B}T/\hbar < \gamma_0 + \beta A/\Omega$.

VII. FALSIFICATION

Three tests can falsify Conjecture 1:

1. *Diffusion-ratio measurement.* At the thermal–quantum boundary (trapped ions, optomechanical oscillators), measure $D_{\rm th}$ and $D_{\rm q}$ independently. Prediction: $R = 2.000 \pm \epsilon$. Rejection: $R \neq 2$ at $> 3\sigma$.
2. *Optimal-erasure friction.* In colloidal Landauer erasure [? ?], the effective damping at quasi-static optimal erasure should satisfy $\gamma_{\rm eff} = k_{\rm B}T/\hbar$. Rejection: $\gamma_{\rm eff} \neq k_{\rm B}T/\hbar$ at the Landauer bound.
3. *Superfluid He-4 second-sound attenuation.* Five-temperature sweep at 0.5–2.17 K. Prediction: $D_{\rm th}/D_{\rm q} = 2$ with $\gamma = k_{\rm B}T/\hbar$. This provides an overconstrained test (5 data points, 1 parameter).

VIII. WHAT IS AND IS NOT PROVEN

Proven: The equivalence of conditions (i)–(v) is exact algebra. The deviation formula (**??**) is exact. The TICE embedding (**??**) with the closure factor (**??**) is exact for any admissible kernel. No approximations beyond the Markov regime for the base theorem, and the kernel framework extends to non-Markovian dynamics.

Not proven: Conjecture 1 ($D_{\rm th} = 2D_{\rm q}$) is not derived from first principles. It is a closure condition motivated by five independent physical observations but not forced by any known axiom. The Bochner positivity of the kernel guarantees *admissibility* (the kernel is a valid bath correlator) but does not *uniquely select* this kernel or this closure. Three routes toward a first-principles derivation exist: (a) a thermofield-double construction proving forward/backward noise independence in open systems; (b) a Guerra–Morato variational argument [?] connecting the Nelson action to Landauer cost; (c) direct experimental measurement of R.

IX. CONCLUSION

A single closure condition—$D_{\mathrm{th}} = 2D_{\mathrm{q}}$, motivated by the bidirectional structure of Nelson's stochastic mechanics—implies $\gamma_0 = k_{\mathrm{B}}T/\hbar$ and a bridge energy of exactly one Landauer bit. Five equivalent formulations, an exact deviation law, a memory-kernel generalization with TICE closure factor, one explicit kernel solution, and three falsification protocols are provided. The result is conditional, and we have stated the single remaining assumption explicitly.

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